UNDERWRITING FORECASTING ANALYSIS

Stage 1 + Stage 2A + Stage 2B

Data validation, behavioural analysis, traditional time-series forecasting and neural forecasting

Transaction Data	**Validation**	**Monthly Series**	**Diagnostics**
ARIMA / SARIMA / ETS	**RNN / LSTM / GRU**	**Error Metrics**	**Final Ranking**

Mentor Review Report

Prepared from the implemented notebooks, verified outputs and submitted plot evidence.

Scope note: ECM/VECM is intentionally excluded. The report covers only the models retained in the implemented forecasting comparison.

1. Executive Summary

This report documents the complete underwriting forecasting workflow implemented from transaction-level data through model comparison. The forecasting target is the monthly credit amount. The workflow validates and enriches transaction data, aggregates it into a monthly time series, examines the statistical behaviour of that series, and then uses those observations to guide model selection.

The forecasting stage contains two modelling paths. Stage 2A uses traditional time-series models: ARIMA, SARIMA and ETS using Holt's double exponential smoothing. Stage 2B uses recurrent neural models: Simple RNN, LSTM and GRU.

All retained models are evaluated against the same held-out June 2026 actual credit value of ₹59,000 using MAE, RMSE and MAPE. The observed ranking places LSTM first at 0.0146% MAPE, followed by ETS at 0.0336%.

The result is a prototype experiment, not a production-level generalisation claim. Only six monthly observations are available, with five used for training and one for testing. The neural path therefore has only two training sequences. This is enough to validate the implementation, but not enough to establish robust long-run model superiority.

2. Project Objective and Scope

The forecasting component is designed to turn historical account behaviour into a reproducible estimate of next-month credit activity. The goal is not only to produce a forecast, but to create a traceable path from raw transactions to a model-backed underwriting signal.

The implementation follows these principles:

- Use transaction-level data as the source of truth.
- Validate the transaction structure before modelling.
- Aggregate raw events into a consistent monthly time series.
- Use statistical diagnostics to guide traditional model selection.
- Use a common held-out observation and common error metrics for model comparison.

The prototype covers six months of prepared transaction data and forecasts the June 2026 monthly credit amount.

3. Complete End-to-End Pathway

Each stage produces an output that becomes the input to the next stage. Traditional models consume the ordered monthly series directly, while neural models consume scaled three-step sequences.

Raw Transactions	**Data Validation**	**Share-Market Flag**	**Monthly Aggregation**
Monthly Credit	**Trend / Moving Average**	**ACF / Stationarity**	**Stage 2A**
ARIMA / SARIMA / ETS	**Forecast**	**MAE / RMSE / MAPE**	**Stage 2A Comparison**
Monthly Credit	**Scaling**	**3-Month Windows**	**RNN / LSTM / GRU**
Neural Forecast	**MAE / RMSE / MAPE**	**Neural Comparison**	**Final Ranking**

Important hand-off: Stage 1 does not directly become a neural-network tensor. Stage 1 produces the cleaned monthly series and statistical understanding. Stage 2A uses the series directly. Stage 2B transforms the same series into sequences suitable for recurrent networks.

4. Stage 1 — Data Preparation and Validation

The starting transaction structure contains four core fields: date, debit, credit and balance. These represent individual account movements. Before forecasting, the records are checked for structural consistency and converted into a regular monthly representation.

4.1 Core fields

Field	Role in the workflow
Date	Defines transaction timing and enables monthly grouping.
Debit	Represents outgoing transaction value.
Credit	Represents incoming transaction value and becomes the primary forecasting target after aggregation.
Balance	Provides account-level consistency context and supports validation of transaction behaviour.

4.2 Share-market behavioural flag

A binary share-market flag was introduced for selected debit transactions. A value of 1 indicates that the debit is treated as related to share-market activity, while 0 indicates that it is not. This is a behavioural feature introduced under the mentor's assumption; it is not the forecast target.

Why it matters

• Captures a class of outgoing financial behaviour relevant to underwriting.

• Separates selected investment-related activity from general spending.

• Can be aggregated into monthly share-market debit amount and share-market debit ratio.

• Enriches the behavioural representation without replacing monthly credit as the forecasting target.

4. Stage 1 — Monthly Aggregation

Transaction records are grouped by month. Credit values are summed to obtain total monthly credit. The same aggregation produces total debit and share-market debit measures.

Month	Total Credit	Total Debit	Share-Market Debit	Share-Market Ratio
2026-01	₹55,000	₹41,984	₹12,480	29.73%
2026-02	₹54,500	₹38,286	₹12,415	32.43%
2026-03	₹57,000	₹41,967	₹12,729	30.33%
2026-04	₹56,000	₹41,930	₹12,638	30.14%
2026-05	₹58,500	₹48,966	₹12,820	26.18%
2026-06	₹59,000	₹39,111	₹12,922	33.04%

Validation evidence

• Dates are treated as temporal keys and converted into monthly periods.

• Credit and debit amounts are treated numerically before aggregation.

• Monthly totals are checked for consistency before forecasting.

• The share-market flag is binary and distinguishes the selected debit transactions.

• The final forecasting target is a six-point monthly credit series.

The resulting monthly credit series is: ₹55,000, ₹54,500, ₹57,000, ₹56,000, ₹58,500 and ₹59,000.

5. Stage 1 — Behavioural and Statistical Analysis

The monthly series is analysed before forecasting because the modelling families make different assumptions. Trend, smoothing, distribution, autocorrelation and stationarity provide the reasoning bridge into Stage 2.

5.1 Key findings

Diagnostic	Observed result	Implication
Trend	Overall upward direction	Supports testing a trend-capable model such as Holt's double exponential smoothing.
3-month moving average	Rises from ₹55,500 to ₹57,833.33	Confirms the smoothed upward direction.
Distribution	W=0.9335, p=0.6075	No strong evidence against normality, but sample is only six points.
ACF lag 1	0.551	Positive nearby-lag dependence.
ACF lag 2	0.860	Strong short-lag relationship; motivated further temporal analysis.
ADF, original	stat0.4248, p=0.9824	No evidence against a unit-root style non-stationarity interpretation.
ADF, first difference	stat7.7068, p1.29×10^{-11}	Differencing strongly changes the stationarity result.

The diagnostic results are descriptive because six observations are an extremely small sample.

5.2 Monthly Credit Trend

The fitted straight-line slope is approximately ₹885.71 per month. The series is not monotonic: February is below January and April is below March, followed by strong increases in May and June.

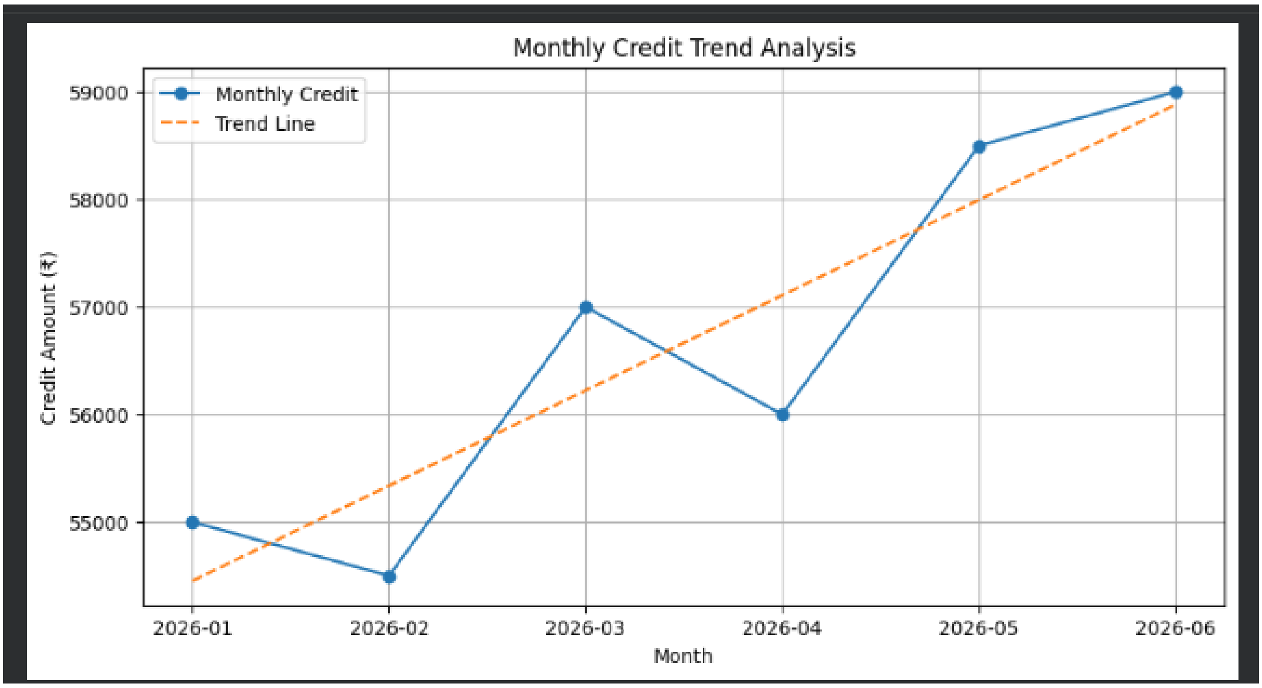

Figure 1. Monthly credit trend with fitted trend line from the submitted notebook.

5.3 Three-Month Moving Average

The moving average values are ₹55,500, ₹55,833.33, ₹57,166.67 and ₹57,833.33 for March through June. The smoothing supports considering a trend-aware model.

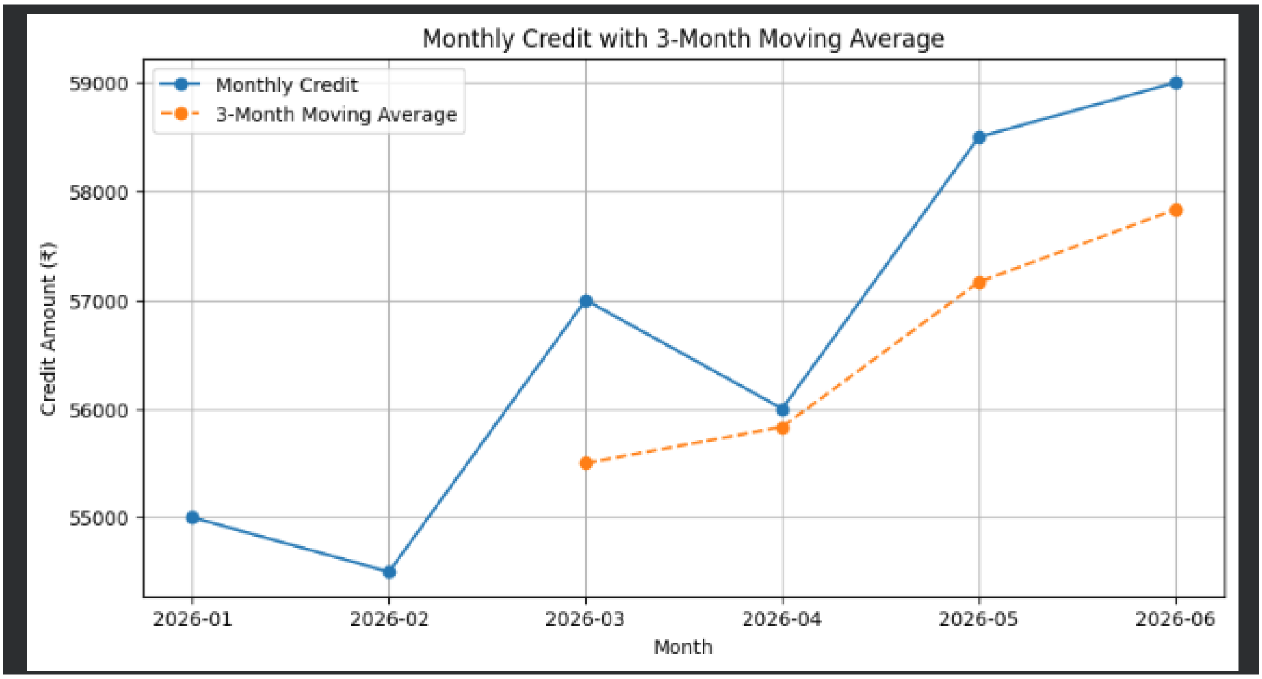

Figure 2. Monthly credit and three-month moving average from the submitted notebook.

5.4 Distribution of Monthly Credit

The six observations lie between ₹54,500 and ₹59,000. Shapiro-Wilk gives W=0.9335 with p=0.6075. This is descriptive only because the sample is very small.

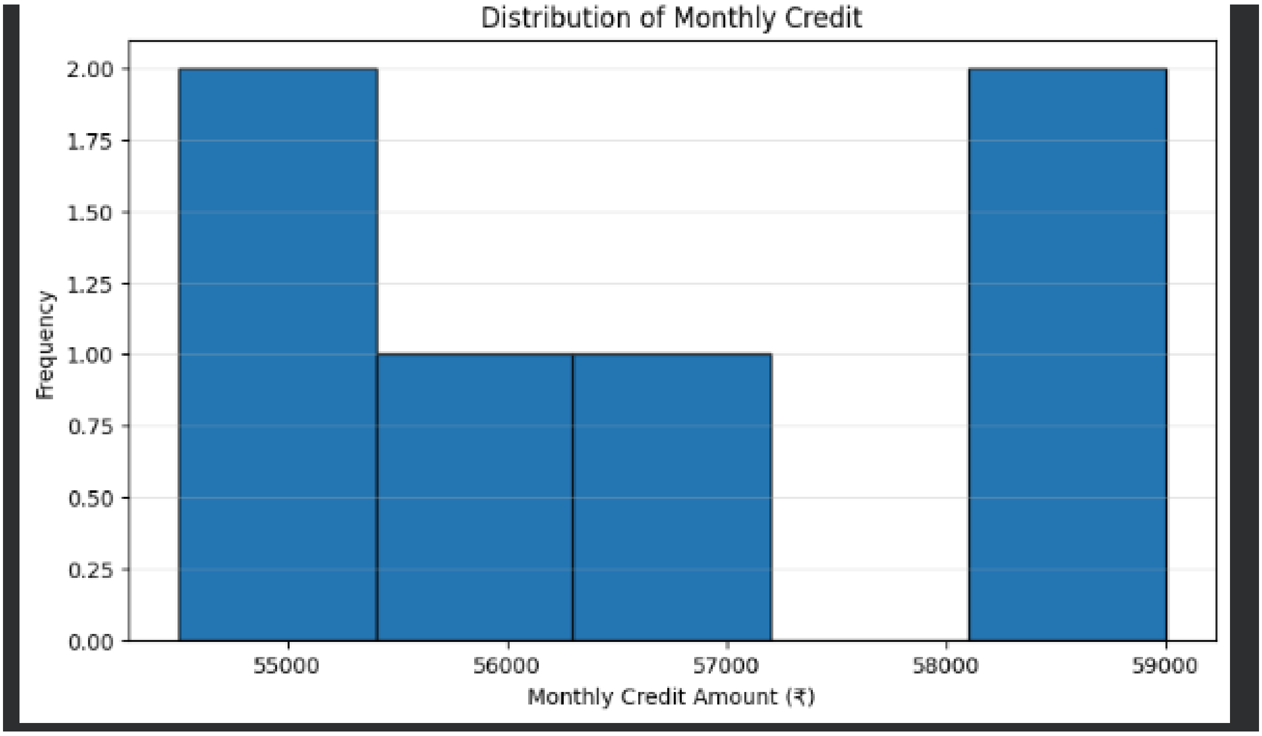

Figure 3. Distribution of monthly credit from the submitted notebook.

5.5 Autocorrelation

The observed autocorrelation is approximately 0.551 at lag 1 and 0.860 at lag 2. Positive nearby-lag relationships support testing autoregressive models.

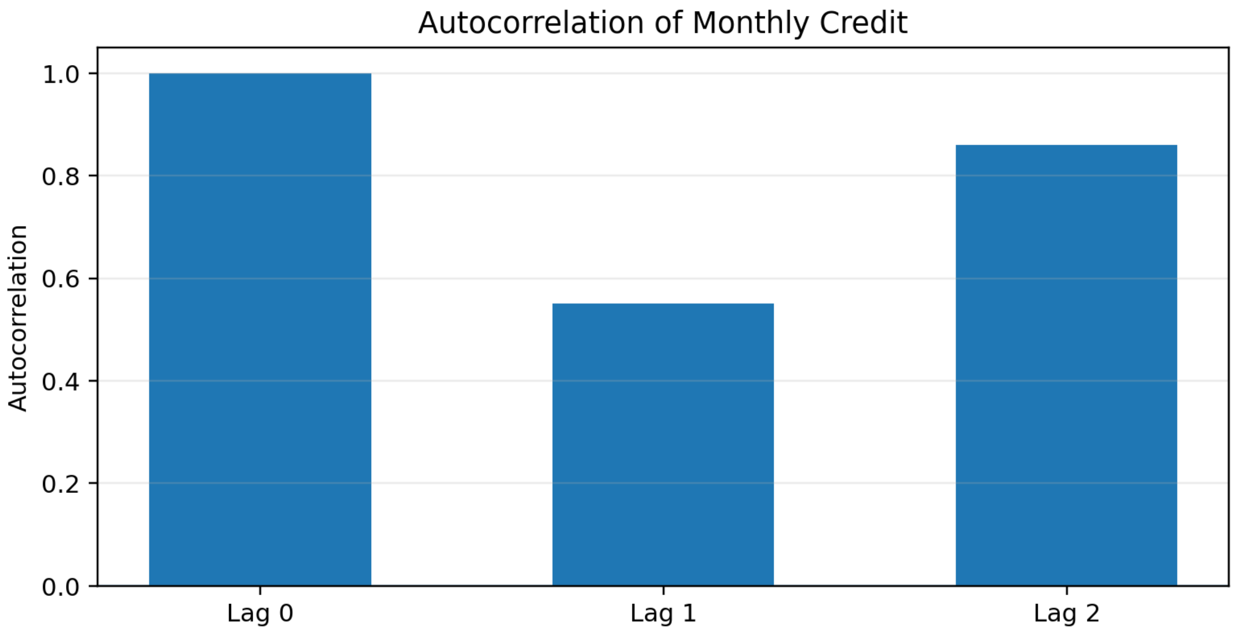

Figure 4. Autocorrelation values derived from the six-month monthly credit series.

5.6 First Differencing

The original ADF result is approximately 0.4248 with $p=0.9824$, while the first-differenced series gives approximately 7.7068 with $p1.29\times10^{-11}$. This is the statistical motivation for differencing in ARIMA-family modelling.

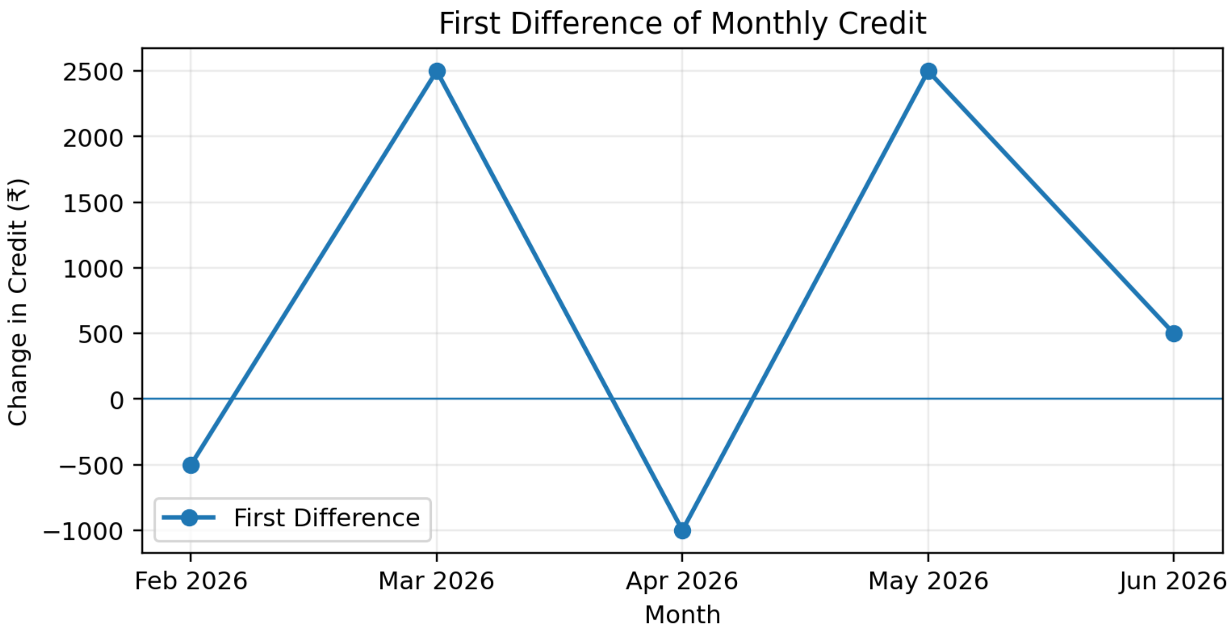

Figure 5. First difference of the monthly credit series.

6. Stage 1 Stage 2 Input Mapping

Different model families use different aspects of the Stage 1 output. The same monthly credit target is retained throughout, while preprocessing changes according to model requirements.

Stage 1 output	Used by	Purpose
Monthly total credit	All models	Primary time-series target.
Observed upward trend	ETS	Supports level + trend smoothing.
Autocorrelation	ARIMA / SARIMA	Supports autoregressive modelling.
Non-stationary level behaviour	ARIMA / SARIMA	Motivates differencing.
Seasonality check	SARIMA	Allows a seasonal extension to be tested.
Scaled monthly values	RNN / LSTM / GRU	Numerically stable neural input.
Three-month windows	RNN / LSTM / GRU	Converts the series into supervised sequences.

Traditional models receive the ordered monthly series. Neural models receive the same series after scaling and conversion into three-step sequences.

7. Stage 2A — Traditional Forecasting

Traditional forecasting provides interpretable statistical baselines. The retained models are ARIMA, SARIMA and ETS using Holt's double exponential smoothing.

Monthly Credit Series | **Statistical Structure** | **Model Fit** | **June Forecast** | **Error Metrics**

Model	Primary input	What it emphasises	Why included
ARIMA(1,1,1)	Monthly credit	Autocorrelation + differencing	Baseline non-seasonal time-series model.
SARIMA(1,1,1)(1,1,1,2)	Monthly credit	Autocorrelation + differencing + seasonal structure	Tests a seasonal extension.
ETS (Holt's Double)	Monthly credit	Level + trend	Directly represents the upward direction observed in Stage 1.

The implemented notebook outputs are used as the source of the numerical results reported below.

7.1 ARIMA(1,1,1)

ARIMA stands for Autoregressive Integrated Moving Average. The implemented configuration is ARIMA(1,1,1): one autoregressive term, one order of differencing and one moving-average term. The model addresses the non-stationary component through differencing and then models the transformed series using past values and past error behaviour.

ARIMA(1,1,1) — June Forecast

Verified result: forecast ₹56,900.91; actual ₹59,000; MAE ₹2,099.09; RMSE ₹2,099.09; MAPE 3.5578%.

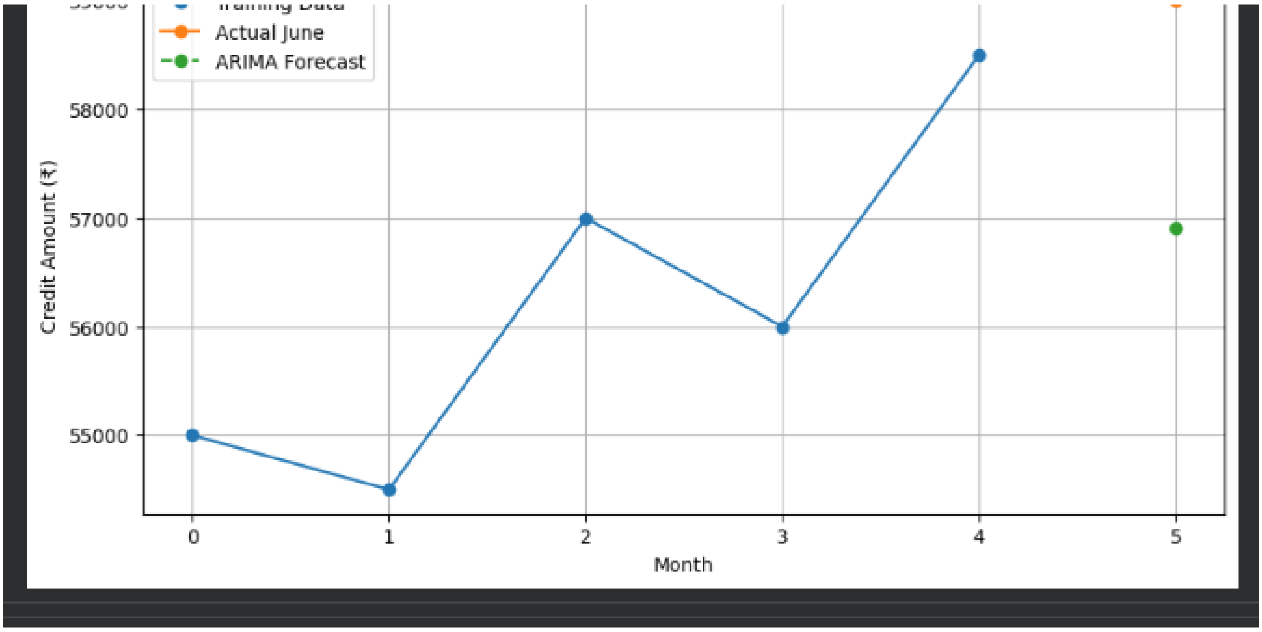

Figure 6. ARIMA(1,1,1) June forecast from the submitted notebook.

7.2 SARIMA(1,1,1)(1,1,1,2)

SARIMA extends ARIMA with seasonal autoregressive, differencing and moving-average terms. The implemented model is SARIMA(1,1,1)(1,1,1,2). The seasonal extension is treated as a comparative experiment rather than proof of a true seasonal cycle because six observations are not enough to establish seasonality reliably.

SARIMA — June Forecast

Verified result: forecast ₹57,500; actual ₹59,000; MAE ₹1,500; RMSE ₹1,500; MAPE 2.5424%.

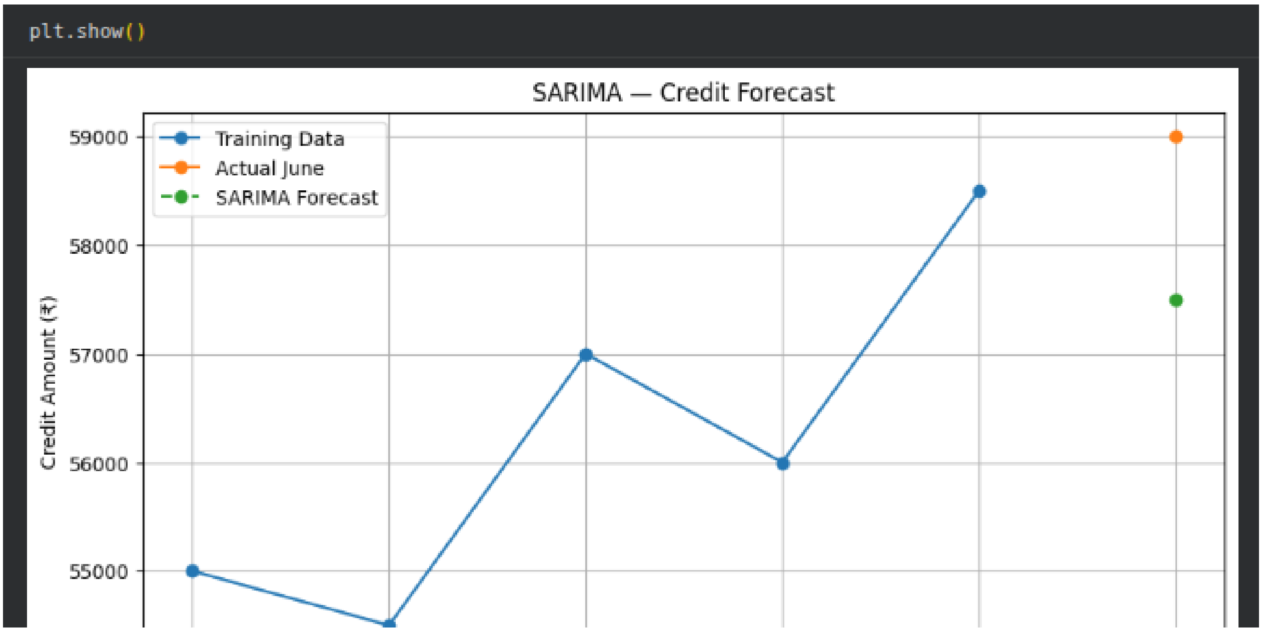

Figure 7. SARIMA June forecast from the submitted notebook.

7.3 ETS — Holt's Double Exponential Smoothing

ETS refers to the Error, Trend and Seasonal framework of exponential smoothing. The implemented model uses Holt's double exponential smoothing, which represents level and trend and gives more weight to recent observations.

ETS (Holt's Double) — June Forecast

Verified result: forecast ₹59,019.84; actual ₹59,000; MAE ₹19.84; RMSE ₹19.84; MAPE 0.0336%.

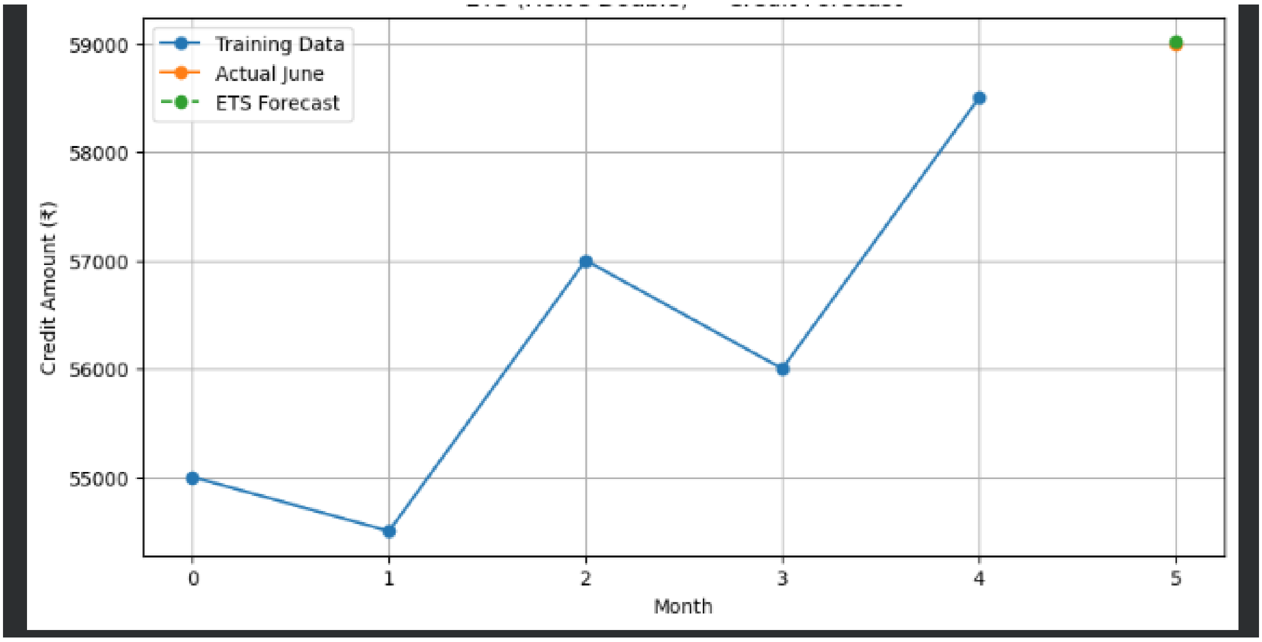

Figure 8. ETS (Holt's Double) June forecast from the submitted notebook.

7.4 Traditional Model Comparison

Model	Forecast	Actual	MAE	RMSE	MAPE
ETS (Holt's Double)	₹59,019.84	₹59,000.00	₹19.84	₹19.84	0.0336%
SARIMA(1,1,1)(1,1,1,2)	₹57,500.00	₹59,000.00	₹1,500.00	₹1,500.00	2.5424%
ARIMA(1,1,1)	₹56,900.91	₹59,000.00	₹2,099.09	₹2,099.09	3.5578%

ETS is the strongest traditional model in this prototype. Its very small error is consistent with the upward direction and the short, trend-dominated series.

Traditional Forecasting — MAPE Comparison

Lower MAPE is better. ETS is substantially closer to the held-out June actual than ARIMA or SARIMA in this run.

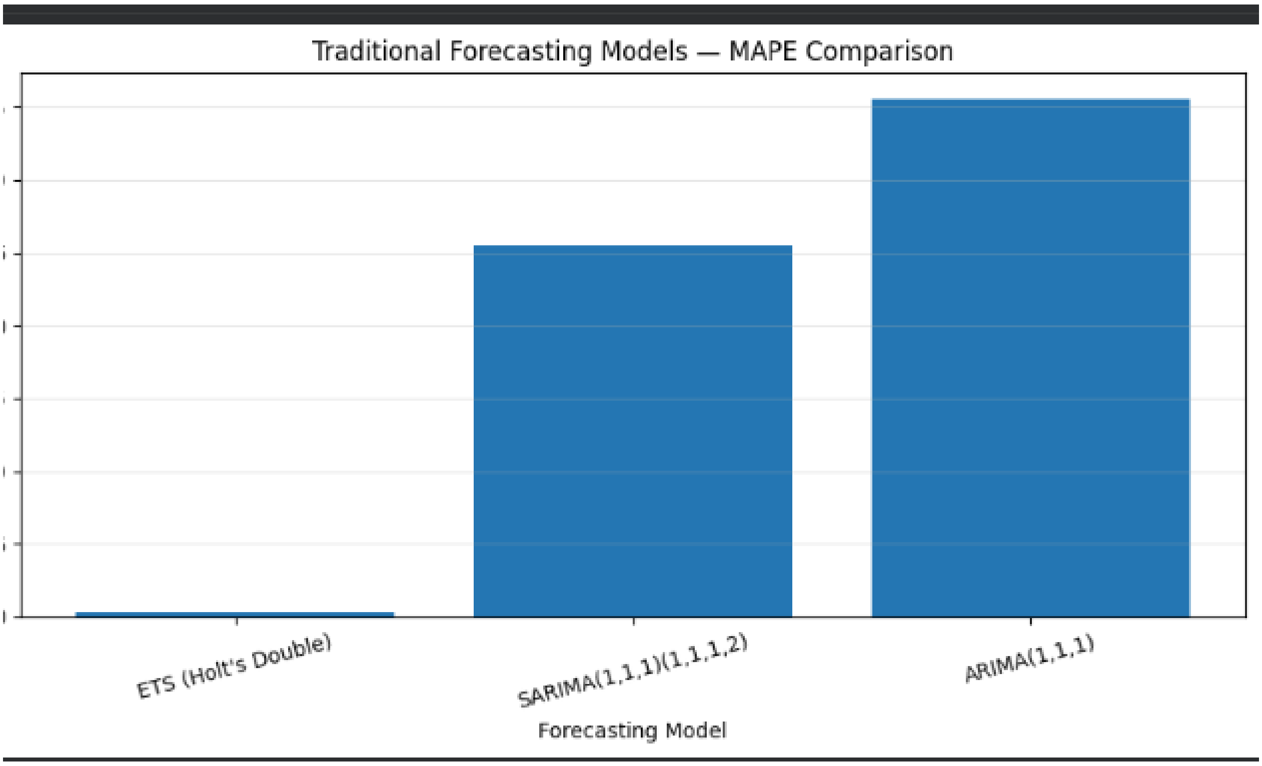

Figure 9. Traditional forecasting MAPE comparison from the submitted notebook.

8. Forecast Error Metrics

The same evaluation metrics are used after both traditional and neural models generate forecasts. This creates a common basis for comparison.

Metric	Meaning	Interpretation
MAE	Mean Absolute Error	Average absolute forecast error in rupees.
RMSE	Root Mean Squared Error	Penalises larger errors more strongly.
MAPE	Mean Absolute Percentage Error	Average absolute percentage error; lower is better when actual values are non-zero.

MAPE is used as the primary ranking metric because it gives a percentage-based comparison. MAE and RMSE are retained to show absolute and squared-error perspectives.

9. Stage 2B — Neural Forecasting

Neural forecasting is a second modelling family rather than a replacement for traditional models. The purpose is to test whether recurrent networks can learn the sequential relationship between recent monthly credit values and the next month's value.

9.1 Common preparation

• Scale the monthly credit values before neural training.

• Create supervised sequences using a three-month input window.

• Reshape the data into the recurrent-network tensor format (samples, timesteps, features).

• Use the same held-out June value and the same error metrics for all three neural models.

With five training months and a window size of three, there are two training sequences. The third window [March, April, May] is the June prediction input. This is a fundamental data-size constraint and should be stated explicitly in any mentor review.

9.2 Tensor shapes

Tensor	Shape	Meaning
X_train	(2, 3, 1)	Two training sequences, three time steps, one feature.
y_train	(2, 1)	Two next-month targets.
X_test	(1, 3, 1)	One held-out three-month sequence for June.

Neural Forecasting Input Pathway

The neural path uses scaling, sequence creation and reshaping before the recurrent layer. These steps are common to RNN, LSTM and GRU.

Figure 10. Neural forecasting pathway used for the recurrent models.

9.3 Simple RNN

The Simple Recurrent Neural Network processes the sequence one time step at a time and carries a hidden state forward. A Dense output layer converts the recurrent representation into one numerical forecast.

Training configuration in the implementation: recurrent layer with a small number of units, Adam optimiser, MSE loss, 200 epochs and batch size 1. The loss curve is monitored to confirm training behaviour.

Simple RNN — Training Loss

The loss drops rapidly in the early epochs and then flattens, showing convergence on the tiny training set. Convergence here does not imply generalisation.

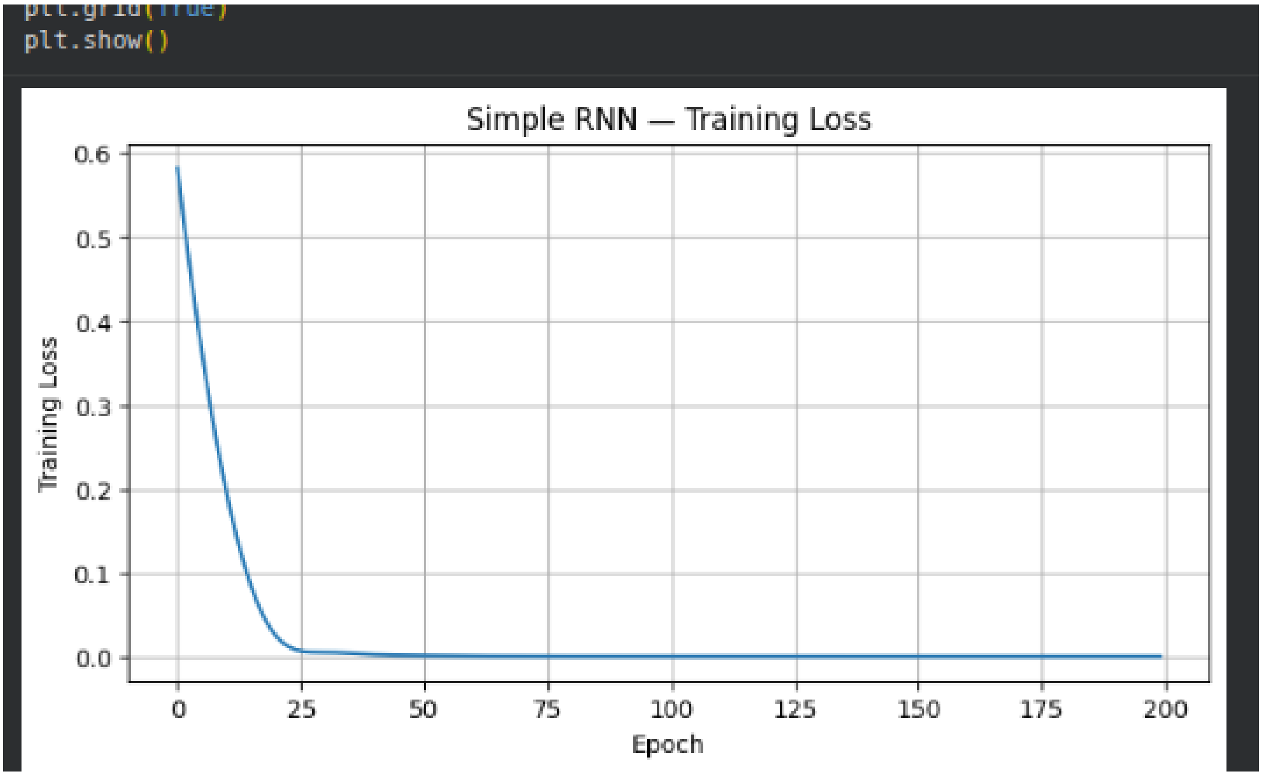

Figure 11. Simple RNN training-loss curve from the submitted notebook.

Simple RNN — June Forecast

Verified result: forecast ₹57,026.42; actual ₹59,000; MAE ₹1,973.58; RMSE ₹1,973.58; MAPE 3.3450%.

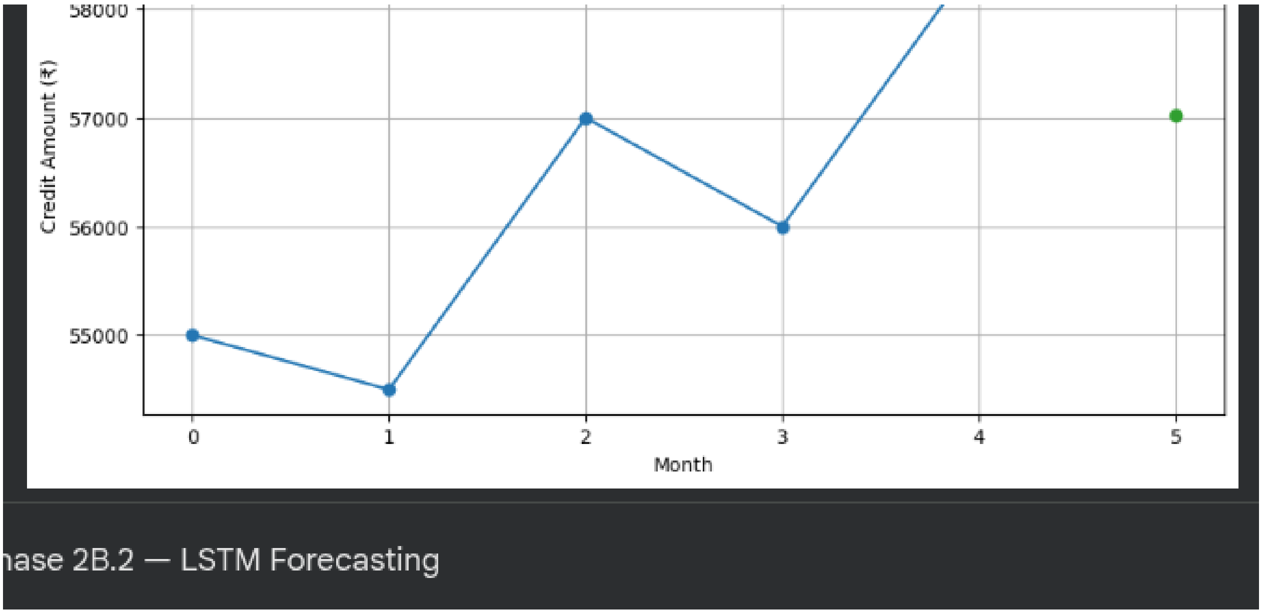

Figure 12. Simple RNN June forecast from the submitted notebook.

9.4 LSTM

LSTM stands for Long Short-Term Memory. It is a gated recurrent architecture that controls what information is retained, updated and exposed across sequence steps. The same three-month input representation is used as in the Simple RNN; the difference is the internal memory mechanism.

LSTM — June Forecast

The standalone LSTM screenshot was not part of the final submitted image set. The figure is therefore a clean reconstruction of the verified result: forecast ₹59,008.61; actual ₹59,000; MAE ₹8.61; RMSE ₹8.61; MAPE 0.0146%.

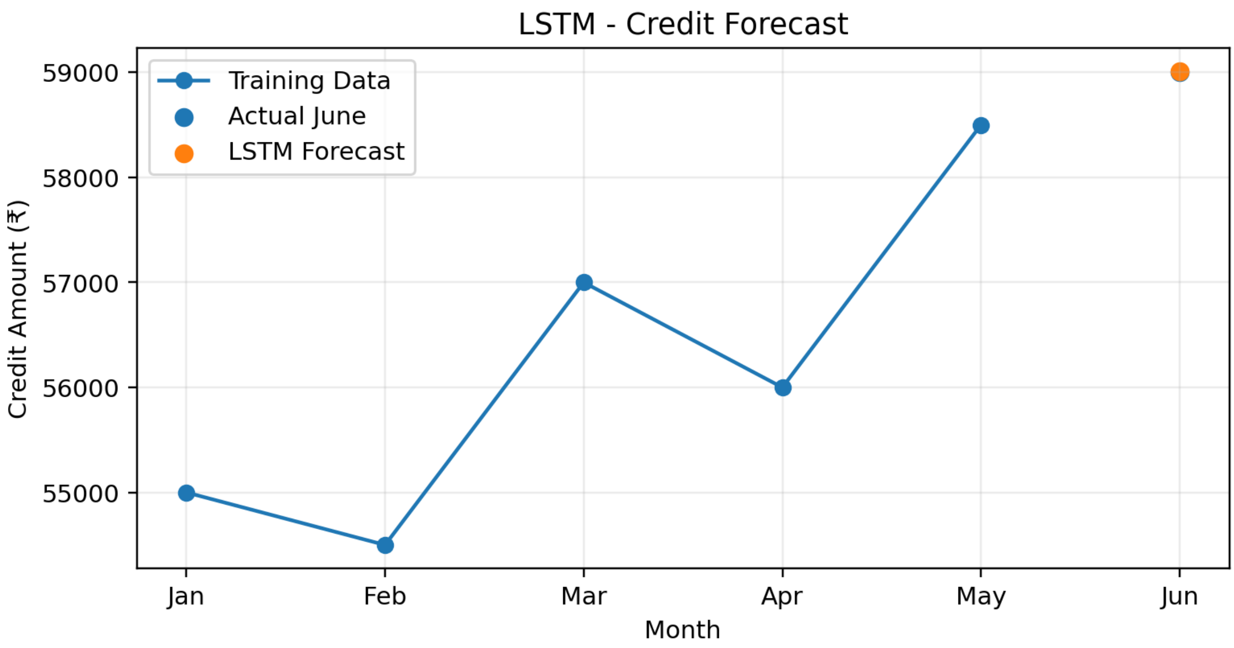

Figure 13. LSTM June forecast reconstructed from the verified notebook result.

9.5 GRU

GRU stands for Gated Recurrent Unit. It uses a simpler gated structure than LSTM. The update gate controls how much old information is retained versus updated, while the reset gate controls how much previous information is considered when forming the new state.

The implemented GRU layer contains 912 parameters and the Dense output layer contains 17 parameters, for 929 trainable parameters.

GRU — June Forecast

The standalone GRU screenshot was not part of the final submitted image set. The figure is therefore a clean reconstruction of the verified result: forecast ₹57,619.36; actual ₹59,000; MAE ₹1,380.64; RMSE ₹1,380.64; MAPE 2.3401%.

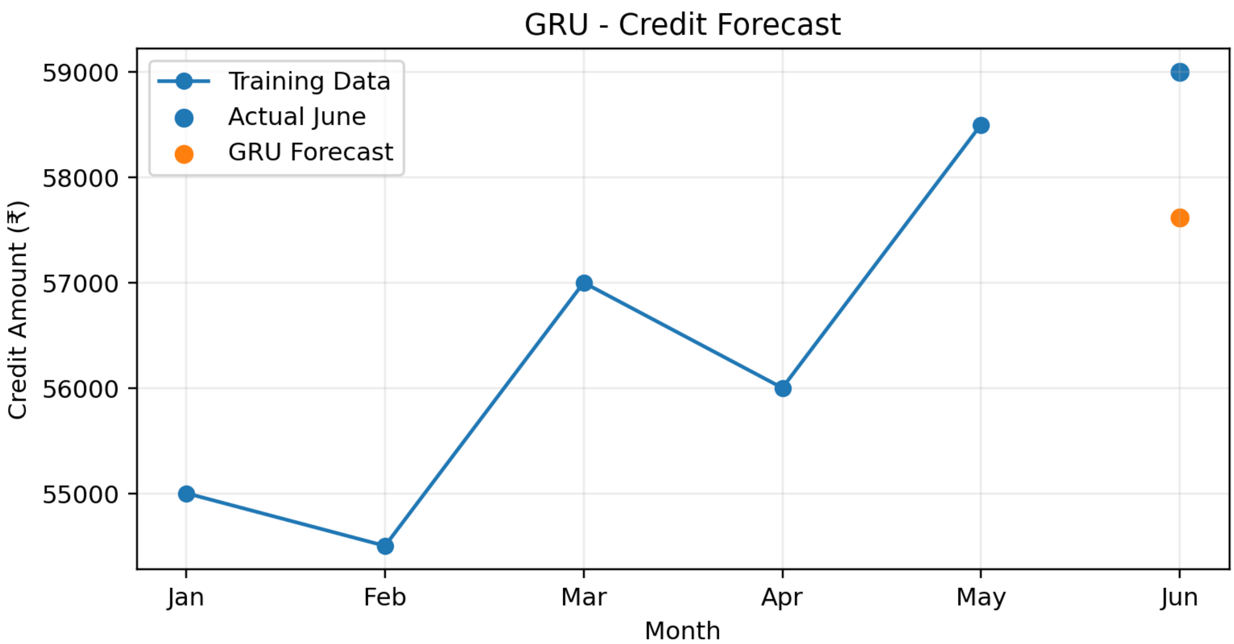

Figure 14. GRU June forecast reconstructed from the verified notebook result.

9.6 Neural Model Comparison

Model	Forecast	Actual	MAE	RMSE	MAPE
LSTM	₹59,008.61	₹59,000.00	₹8.61	₹8.61	0.0146%
GRU	₹57,619.36	₹59,000.00	₹1,380.64	₹1,380.64	2.3401%
Simple RNN	₹57,026.42	₹59,000.00	₹1,973.58	₹1,973.58	3.3450%

LSTM has the lowest observed error. GRU performs better than Simple RNN in this run, but the six-month dataset is too small for a general architectural conclusion.

Neural Forecasting — MAPE Comparison

LSTM is the best neural model in this experiment, with a MAPE of 0.0146%.

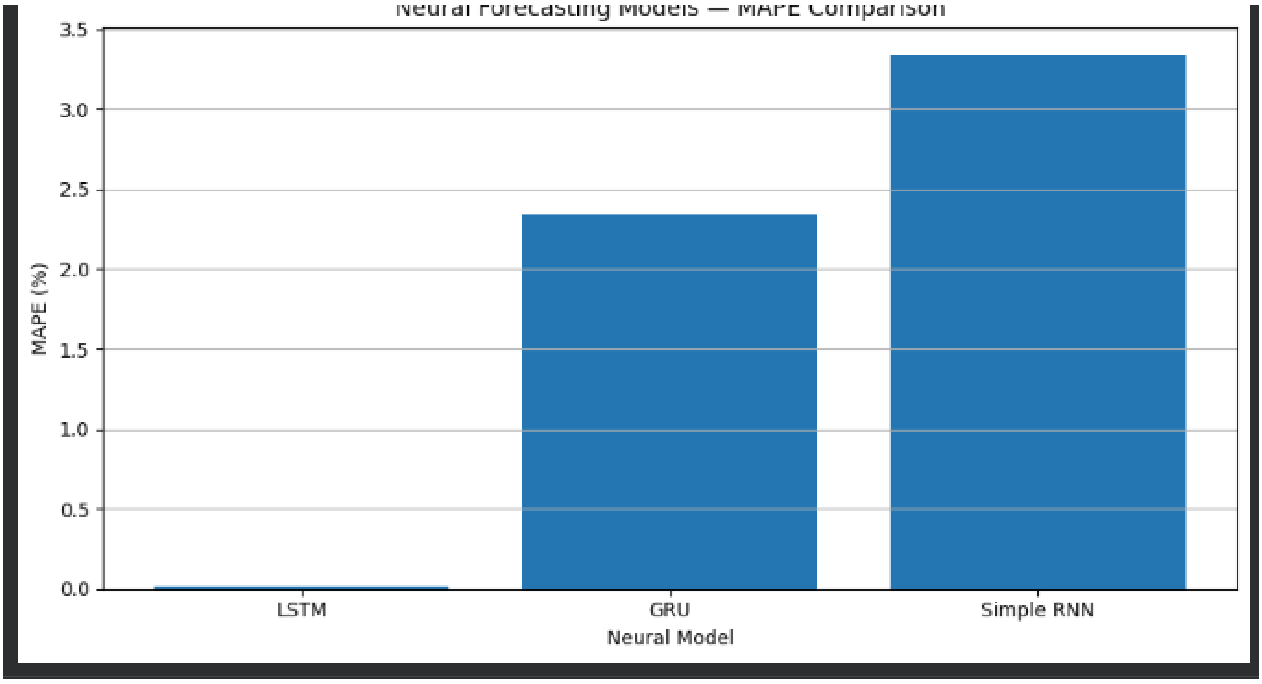

Figure 15. Neural forecasting MAPE comparison from the submitted notebook.

10. Traditional vs Neural Comparison

The two modelling families can be compared because they use the same held-out June value and the same error definitions.

Rank	Model	Family	MAPE
1	LSTM	Neural	0.0146%
2	ETS (Holt's Double)	Traditional	0.0336%
3	GRU	Neural	2.3401%
4	SARIMA(1,1,1)(1,1,1,2)	Traditional	2.5424%
5	Simple RNN	Neural	3.3450%
6	ARIMA(1,1,1)	Traditional	3.5578%

The observed ranking is led by LSTM at 0.0146% MAPE, followed by ETS at 0.0336%. LSTM's forecast differs from the actual June value by ₹8.61, while ETS differs by ₹19.84.

Final Traditional vs Neural Comparison

The final comparison shows that LSTM produced the closest June forecast in this particular prototype, while ETS is the strongest traditional baseline.

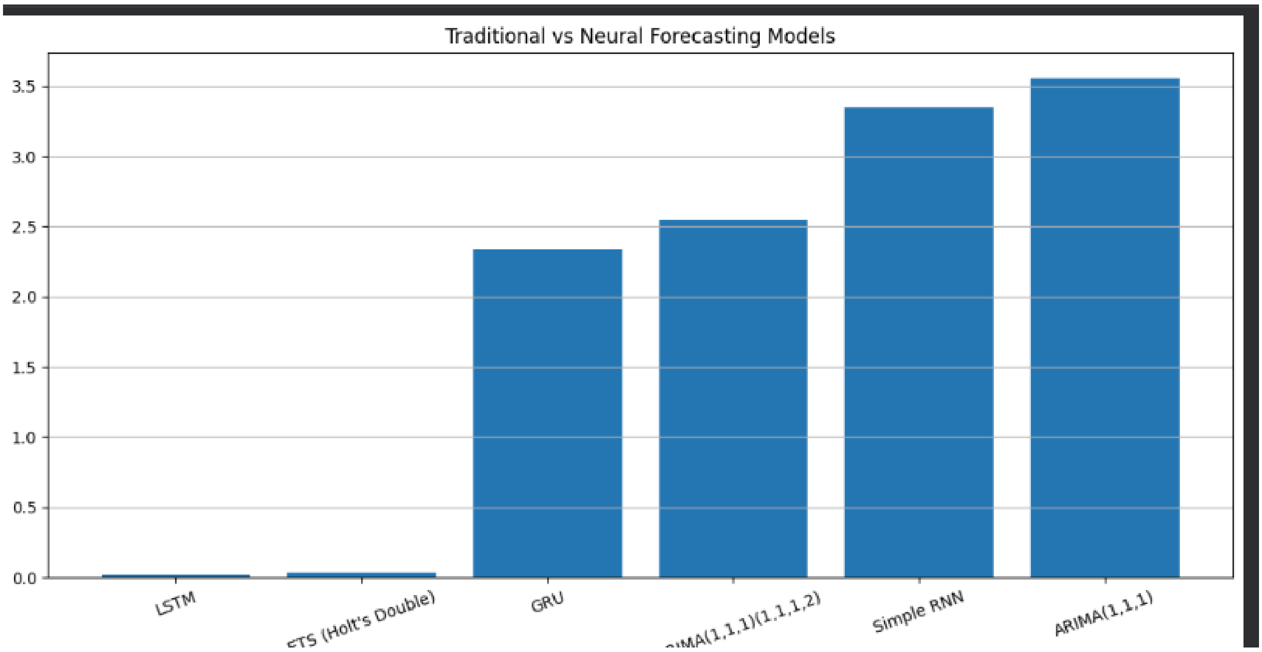

Figure 16. Final traditional versus neural MAPE comparison from the submitted notebook.

11. Model Selection Summary

Model	Stands out for	Main limitation in this prototype
ARIMA	Autocorrelation + differencing	Sensitive to very small samples and statistical specification.
SARIMA	Seasonal extension	Six observations are insufficient to establish seasonality strongly.
ETS	Level + trend	Less expressive for complex nonlinear relationships.
Simple RNN	Basic sequence learning	Very little data and possible difficulty retaining information.
LSTM	Controlled memory	More complex and data-hungry.
GRU	Efficient gated recurrence	May not match LSTM under every dataset or configuration.

The modelling progression is deliberate: establish statistical baselines, inspect error behaviour, then test recurrent neural alternatives under the same held-out evaluation.

12. Model Input and Processing Summary

Model	Input	Processing	Output
ARIMA	Monthly credit series	AR + differencing + MA	Next-month credit
SARIMA	Monthly credit series	ARIMA + seasonal terms	Next-month credit
ETS	Monthly credit series	Level + trend smoothing	Next-month credit
Simple RNN	3-month sequence	Hidden-state recurrence	Next-month credit
LSTM	3-month sequence	Gated memory	Next-month credit
GRU	3-month sequence	Update + reset gates	Next-month credit

The forecast-generation mechanism changes across models, but the evaluation stage is common: predicted value versus actual value, followed by MAE, RMSE and MAPE.

13. Implementation Evidence Record

The submitted notebook screenshots provide direct evidence for the executed diagnostics, traditional forecasts, neural training and model comparisons.

Evidence	What it confirms
Monthly credit trend plot	Monthly series and upward direction were visualised.
Three-month moving average	Smoothing calculation was executed.
Distribution plot	Monthly credit distribution was visualised.
ARIMA forecast plot	ARIMA(1,1,1) was executed and visualised.
SARIMA forecast plot	SARIMA was executed and visualised.
ETS forecast plot	Holt's double exponential smoothing was executed and visualised.
Traditional MAPE plot	Traditional model errors were compared.
Simple RNN loss plot	Neural training loss was monitored.
Simple RNN forecast plot	RNN forecast was compared with actual June.
Neural MAPE plot	RNN, LSTM and GRU errors were compared.
Final comparison plot	All retained models were ranked using MAPE.

Evidence note: the LSTM and GRU standalone forecast figures in this report are clean reconstructions of verified notebook results because those two standalone screenshots were not included in the final submitted image set. This is stated explicitly rather than presenting reconstructed graphics as original screenshots.

14. Prototype Limitations and Interpretation

• Only six monthly observations are available.

• Only five months are used for fitting and one month is held out for testing.

• The neural path has only two training sequences with a three-month window.

• A single held-out month cannot provide a robust long-run estimate of forecasting performance.

• Neural results can be sensitive to initialization and optimisation settings, especially with tiny datasets.

• Seasonality is difficult to establish from six observations, so the SARIMA seasonal extension should be interpreted as a comparison experiment.

• The share-market flag is currently an assumed behavioural classification and should be strengthened with verified labels or robust transaction-description rules in production.

• The current forecasting target is monthly total credit. The additional behavioural features are available for future extensions but are not used as multivariate predictors in this comparison.

The strongest supported conclusion is that LSTM produced the lowest observed June forecast error in this prototype, while ETS produced the strongest traditional-model result. Both should be re-evaluated on a larger historical dataset before a production choice is made.

15. Research Foundation and Related Work

Box, G. E. P. and Jenkins, G. M. (1970). Time Series Analysis: Forecasting and Control.
Foundational work for Box–Jenkins time-series modelling and ARIMA-family model identification, estimation and checking.
Reference: Google Books

Dickey, D. A. and Fuller, W. A. (1979). Distribution of the Estimators for Autoregressive Time Series with a Unit Root.
Foundation for unit-root testing and the statistical motivation behind stationarity analysis.
Reference: Journal of the American Statistical Association / DOI 10.1080/01621459.1979.10482531

Holt, C. C. Work on forecasting trends using exponentially weighted moving averages.
Foundation for exponential-smoothing approaches and trend-aware forecasting.
Reference: International Journal of Forecasting / DOI 10.1016/j.ijforecast.2003.09.015

Hochreiter, S. and Schmidhuber, J. (1997). Long Short-Term Memory. Neural Computation, 9(8), 1735–1780.
Original LSTM paper describing the gated memory mechanism designed to address long-term dependency problems in recurrent learning.
Reference: DOI 10.1162/neco.1997.9.8.1735

Cho, K. et al. (2014). Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation.
Important early work associated with the gated recurrent unit architecture used as the GRU comparison family.
Reference: ACL Anthology, D14-1179

Hyndman, R. J. and Koehler, A. B. (2006). Another look at measures of forecast accuracy. International Journal of Forecasting, 22(4), 679–688.
Research foundation for comparing forecast-error measures and understanding their limitations.
Reference: DOI 10.1016/j.ijforecast.2006.03.001

These references provide methodological foundations for the model families and evaluation measures. They do not turn the six-month prototype into a statistically conclusive experiment.

16. Final Conclusion

The implemented underwriting forecasting pipeline connects transaction-level preparation with statistical diagnostics, traditional forecasting and neural forecasting. The workflow is modular: transaction validation produces a monthly representation; Stage 1 explains its behaviour; Stage 2A establishes statistical baselines; Stage 2B tests recurrent neural alternatives; and the final comparison evaluates all retained models using the same held-out observation and error metrics.

The observed final ranking is: LSTM 0.0146% MAPE, ETS 0.0336%, GRU 2.3401%, SARIMA 2.5424%, Simple RNN 3.3450%, and ARIMA 3.5578%.

The result demonstrates why both modelling families are useful. Traditional models provide transparent statistical baselines and can perform extremely well on short, trend-dominated series. Neural models provide a flexible sequence-learning alternative, but their advantage becomes more meaningful when substantially more historical observations and richer behavioural features are available.

The next production-oriented evolution should focus on expanding historical data, validating the share-market classification, adding justified behavioural predictors, using rolling or walk-forward evaluation, and repeating the comparison across multiple forecast horizons rather than relying on one held-out month.

END OF REPORT